📘 **AI/ML Internship – Day 58 Notes & Tasks**

🟢 **Module 8: Introduction to Chatbot Development**

📌 Part 1: Recap of Day 57

Yesterday we learned:

✅ Pre-trained Word Embeddings

✅ GloVe

✅ FastText

✅ Word Representations

Today we will start a new module:

🤖 **Module 8: Chatbot Development**

Today we will understand:

- What is a Chatbot?
- How does a chatbot work?
- Types of chatbots
- Real-world applications
- Chatbot architecture

📌 Part 2: What is a Chatbot?

Definition

A chatbot is an Artificial Intelligence (AI) program that communicates with humans through text or voice.

It can answer questions, solve problems, provide information, and perform tasks automatically.

Simple Definition

A chatbot is a virtual assistant that can understand user messages and respond automatically.

Examples

- ChatGPT
- Google Assistant
- Siri
- Alexa
- Banking Chatbots
- Customer Support Bots

📌 Part 3: Real-Life Example

Suppose you visit an online shopping website.

You see:

👋 Hi!
How can I help you today?

You ask:

Where is my order?

Bot replies:

Please enter your order number.

You enter:

12345

Bot replies:

Your order has been shipped.

No human is involved.

📌 Part 4: Why Do We Need Chatbots?

Businesses receive thousands of customer questions daily.

Example:

- Order status

- Payment issues
- Refund requests
- Product information
- Business timings

Hiring humans for every query is expensive.

Chatbots can:

- Work 24×7
- Reply instantly
- Handle multiple users
- Reduce company costs

📌 Part 5: How Does a Chatbot Work?

A chatbot follows these basic steps:

User Message
↓
Understand the Message
↓
Find the User's Intent
↓
Search for the Best Response
↓
Reply to the User

Example

User:

I forgot my password

Chatbot understands:

Intent:

Password Reset

Response:

Click "Forgot Password" to reset your password.

📌 Part 6: Chatbot Architecture

```
        User
          |
          ▼
     User Message
          |
          ▼
   Natural Language Processing (NLP)
          |
          ▼
    Intent Recognition
          |
          ▼
 Decision & Response Generator
          |
          ▼
     Response to User
```

📌 Part 7: Components of a Chatbot

1. User Interface (UI)

Where the user types messages.

Examples:

- Website
- Mobile App
- WhatsApp
- Telegram

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2. NLP Engine

Processes the user's text.

Example:

User:

Book a ticket to Delhi

The NLP engine understands that the user wants to book a ticket.

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3. Intent Recognition

Determines the user's purpose.

Example:

Message:

I want to cancel my order

Intent:

Cancel Order

---

4. Response Generator

Generates the appropriate response.

Example:

Please provide your order ID.

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5. Database or Knowledge Base

Stores information that the chatbot uses to answer questions.

Examples:

- Product details
- FAQs
- Customer records

📌 **Part 8: Types of Chatbots**

1. Rule-Based Chatbot

Works using predefined rules.

Example:

User:

Hi

Bot:

Hello!

User:

Bye

Bot:

Goodbye!

Cannot answer questions outside its programmed rules.

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2. AI-Based Chatbot

Uses Artificial Intelligence and Machine Learning.

Can understand different ways of asking the same question.

Example:

Where is my package?

Track my order.

Has my order been shipped?

The chatbot understands that all three have the same intent.

📌 **Part 9: Rule-Based vs AI Chatbot**

Rule-Based	AI-Based
Uses predefined rules	Learns from data
Limited responses	Flexible responses
No learning	Learns continuously
Easy to build	More complex
Suitable for FAQs	Suitable for smart assistants

📌 Part 10: NLP in Chatbots

Natural Language Processing helps chatbots understand human language.

Tasks include:

- Tokenization
- Stop Word Removal
- Lemmatization
- Intent Detection
- Entity Recognition

Example:

Message:

Book a train ticket to Mumbai tomorrow.

Intent:

Book Ticket

Entities:

Destination: Mumbai

Date: Tomorrow

📌 Part 11: Real-World Applications

Customer Support

Answer customer queries.

Banking

- Balance enquiry
- Fund transfer
- Loan information

Healthcare

- Appointment booking
- Symptom checking

Education

- Student doubt clearing
- Course information

E-commerce

- Product search
- Order tracking
- Return requests

📌 **Part 12: Advantages of Chatbots**

✅ Available 24×7

✅ Instant replies

✅ Cost-effective

✅ Handles multiple users

✅ Consistent responses

📌 **Part 13: Limitations**

❌ Cannot understand every question perfectly

❌ Needs training data

❌ May fail with complex conversations

❌ Depends on good NLP models

📌 **Part 14: Popular Chatbot Platforms**

- ChatGPT
- Google Dialogflow
- Microsoft Bot Framework
- Amazon Lex
- IBM Watson Assistant
- Rasa

📌 Part 15: Chatbot Development Workflow

Collect User Requirements
↓
Design Conversation Flow
↓
Prepare Training Data
↓
Train NLP Model
↓
Build Chatbot
↓
Test Chatbot
↓
Deploy
↓
Collect User Feedback
↓
Improve Chatbot

📌 Part 16: Theory Questions

What is a chatbot?

A chatbot is an AI application that communicates with users through text or voice.

What are the two types of chatbots?

- Rule-Based Chatbot
- AI-Based Chatbot

What is Intent?

The purpose or goal behind a user's message.

Why is NLP important in chatbots?

NLP helps chatbots understand human language and respond appropriately.

Give three applications of chatbots.

- Customer Support
- Banking
- Healthcare

📌 Part 17: Interview Questions

Explain chatbot architecture.

A chatbot receives a user message, processes it using NLP, identifies the user's intent, generates a response, and sends it back.

Difference between Rule-Based and AI Chatbots.

Rule-based chatbots follow predefined rules, while AI chatbots use NLP and machine learning to understand user intent.

What is Intent Recognition?

It is the process of identifying the user's goal from a message.

Why are chatbots used in businesses?

To provide 24/7 support, reduce costs, and improve customer experience.

Name some chatbot development platforms.

ChatGPT, Dialogflow, Rasa, Amazon Lex, IBM Watson Assistant.

🎯 Tasks for Day 58

🧠 Theory

1. Define a chatbot.
2. Explain chatbot architecture.
3. What is NLP in chatbots?
4. Differentiate Rule-Based and AI-Based chatbots.
5. List five real-world applications of chatbots.

💻 Practical Task 1

Draw the chatbot architecture diagram.

💻 Practical Task 2

List five websites or applications that use chatbots.

💻 Practical Task 3

Write five customer questions and identify their intents.

Example:

User Message	Intent
Where is my order?	Track Order
I forgot my password	Password Reset

💻 Practical Task 4

Create a simple conversation flow for a banking chatbot.

💻 Practical Task 5

Compare Rule-Based and AI-Based chatbots in a table.